* What is Agentic AI

→ Agentic AI is a type of AI that can take up a task or goal from a user and then work toward completing it on its own, with minimal human guidance. It plans, takes action, adapts to changes, and seeks help only when necessary.

Gen AI	Agentic AI
↓	↓
Reactive (to the point response)	Proactive

Key characteristics

Agentic AI

- Autonomous
- Goal Oriented
- Planning
- Reasoning
- Adaptability
- Context Awareness

Autonomy

→ Autonomy refers to the AI system's ability to make decisions and take actions on its own to achieve a given goal, without needing step-by-step human instructions.

↳ Our AI recruiter is autonomous

↳ It's proactive

→ Autonomy can be controlled

↳ Permission scope

↳ Human-in-the-Loop (HITL)

↳ Override controls

↳ Guardrails/Policies

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Goal Oriented

means that the AI system operates with a Persistant objective in mind and continuously directs its actions to acheive that objective, rather than just responding to isolated Prompts.

→ Goals acts as a compass for autonomy
→ Goals can come with constrains
→ Goals are stored in core memory
→ Goals can be altered.

Planning

→ is the agent's ability to break down a high level goal into a structured sequence of actions or subgoals and decide the best path to acheive the desired outcome.

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Step 1: Generating multiple candidate plans

Step 2: Evaluate each plan

Step 3: Select the best plan with the help of

Reasoning

Reasoning is the cognitive process through which an agentic ai system interprets information, draws conclusions, and make decisions - both while planning ahead and while executing actions in real time.

→ Goal decomposition

→ Tool selection

→ Resource estimation

During Execution:

→ Decision-making

→ HITL handling

→ Error handling

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Adaptability

Adaptability is the agent's ability to modify its plans, strategies, or actions in response to unexpected conditions – all while staying aligned with the goal.

→ Failures (calender API)
→ External feedback (less no of applications)
→ changing goals (Hiring a freelancer)

Context Awareness

Context awareness is the agent's ability to understand, retain, and utilize relevant information from the ongoing task, past interaction, user preferences, and environmental cues to make better decisions throughout a multi-step process.

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→ Types of context

↳ The original goal

↳ progress till now + Interaction history

↳ Environment state

↳ Tool response

↳ User specific preferences

↳ Policy or Guardrails

→ Context awareness is implemented through memory

→ Short term memory

→ Long term memory

Components

- Brain
- Orchestrator
- Tools
- Memory
- Supervisor

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Brain

→ LLM based

↳ Goal Interpretation

↳ Planning

↳ Reasoning

↳ Tool Selection

↳ communication (NLP)

Orchestrator (like Manager)

→ Planning Executes

↳ Task sequencing

↳ conditional Routing

↳ Retry logic

↳ Looping & Iteration

↳ Delegation

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Tools

→ External Action (Perform API calls)

→ Knowledge Base Access (Retrieve factual using RAG)

Memory

→ short-term memory

→ Long-term memory

→ State Tracking

Supervision

(Agent + Human)

→ Approval Request (HITL)

→ ~~Guard~~ Guardrails Enforcement

→ Edge case Escalation

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